



German Conference on Bioinformatics

WS8: Spatial domain identification: computational methods for discovering tissue architecture

Robin Khatri
Institute of Medical Systems Bioinformatics
University Medical Center Hamburg-Eppendorf
21.09.2025



ZMNH
Zentrum für Molekulare
Neurobiologie Hamburg



Universitätsklinikum
Hamburg-Eppendorf

Part I

Methods

Methods and datasets

GE + XY

GraphST
ConST
STAGATE
Spatial-MGCN
STMGCN

SEDR
SCGDL
SpaceFlow
CCST
spaGCN

Methods
($n = 19$)

Real datasets
($n = 30$)

Synthetic datasets
($n = 27$)

GE

Leiden
Louvain
Seurat
BayesSpace

GE + H&E

Muse
Spacell

GE + XY + H&E

stLearn
DeepST
SpaGCN+

Identifying spatial domains



Domain 1
Domain 2
Domain 3
Domain 4
Domain 5

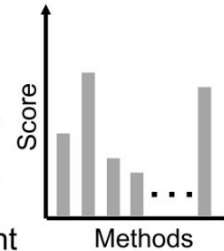
▮ Accuracy

▮ Stability

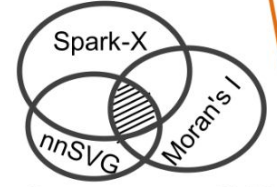
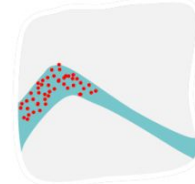
▮ Generality

▮ Scalability

▮ Refinement



Detecting domain-specific SVGs



Consensus SVGs

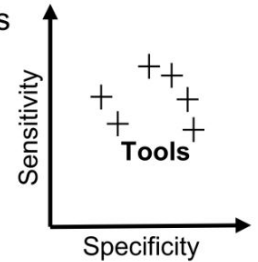
▮ No. of SVGs

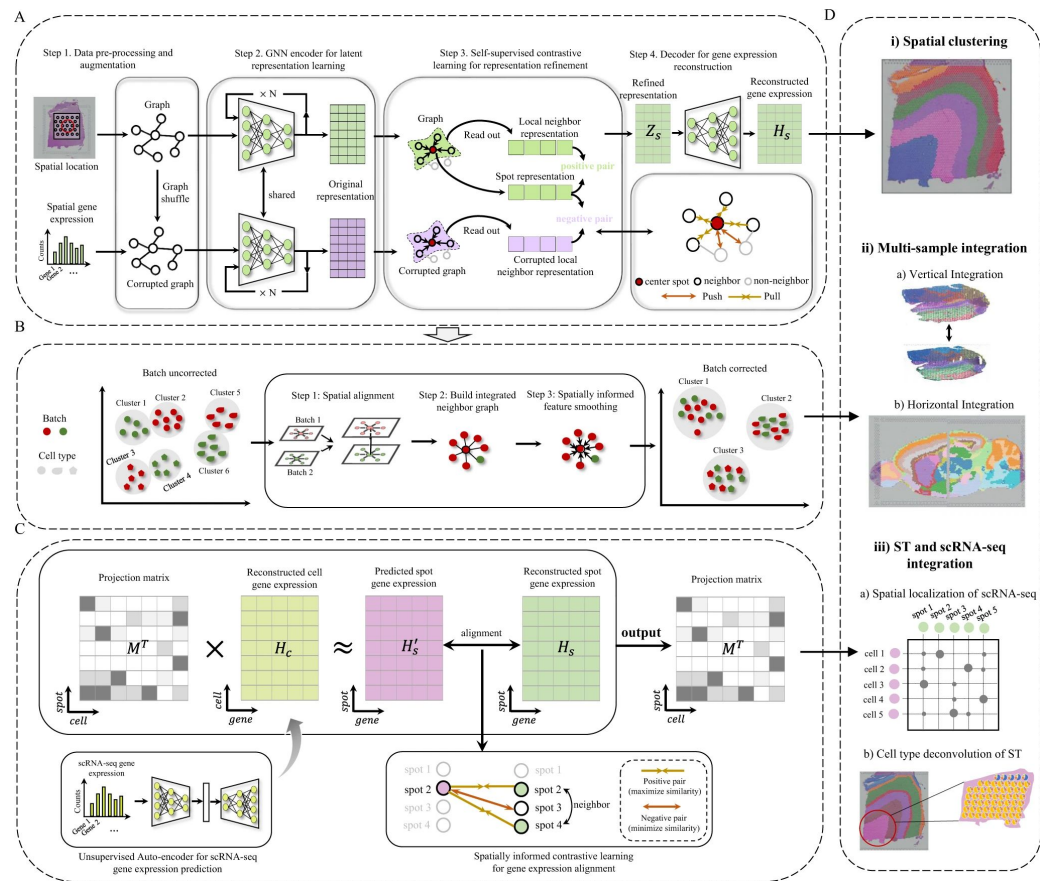
▮ Moran's I

▮ Geary's C

▮ AUPR

▮ SN / SP



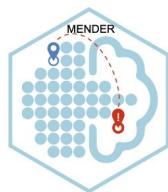
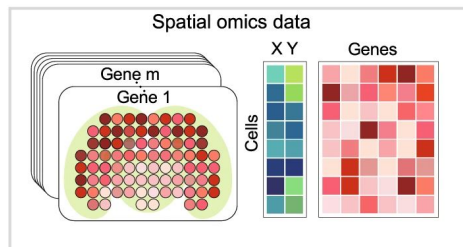


GraphST

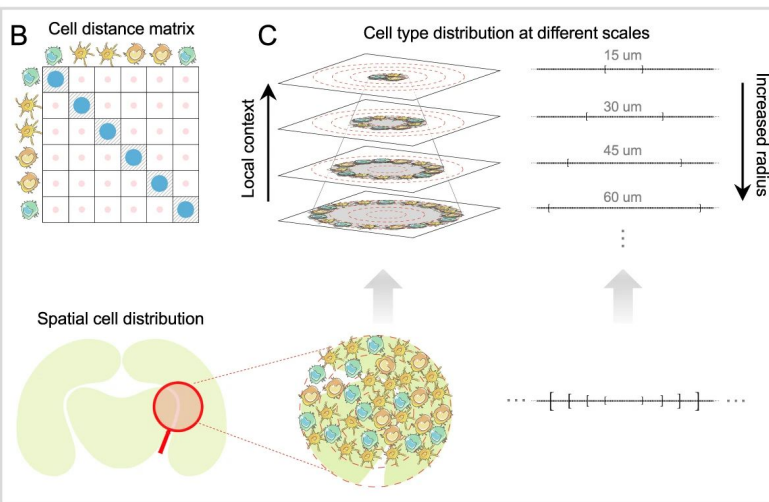
Long et al., 2023

<https://academic.oup.com/nar/article/53/7/gkaf303/8114322>

A

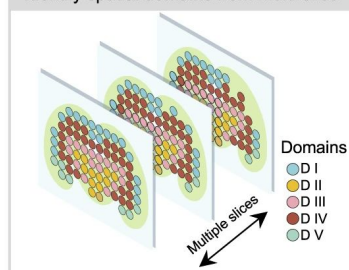


MENDER
Multi-range cEl
coNtext DEcipherER



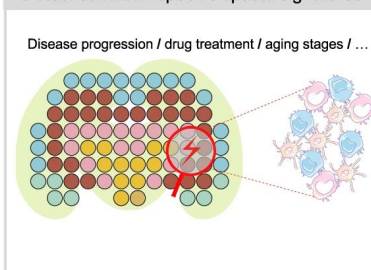
D

Identify spatial domains from multi-slice



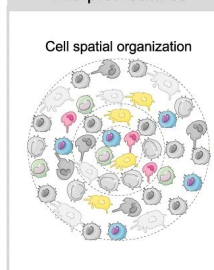
E

Detect condition-specific spatial signatures



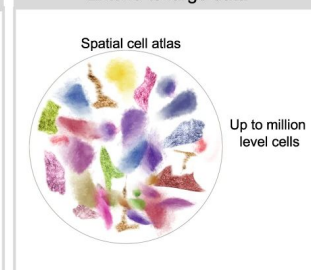
F

Interpret features



G

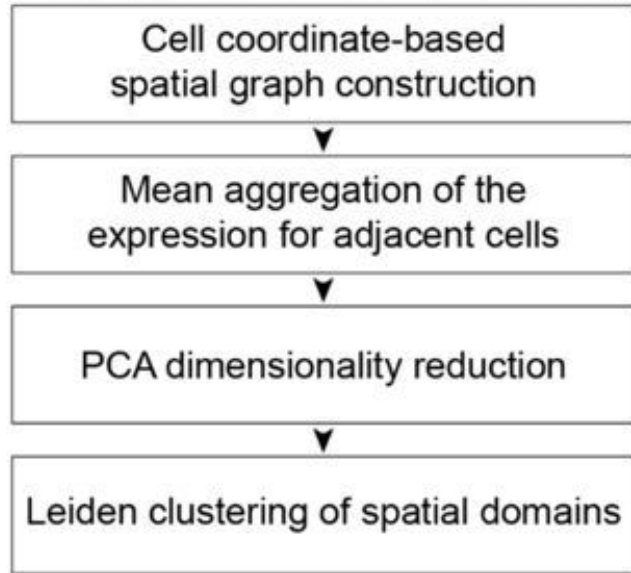
Extend to large data



MENDER

Yuan, 2024

<https://www.nature.com/articles/s41467-023-44367-9>



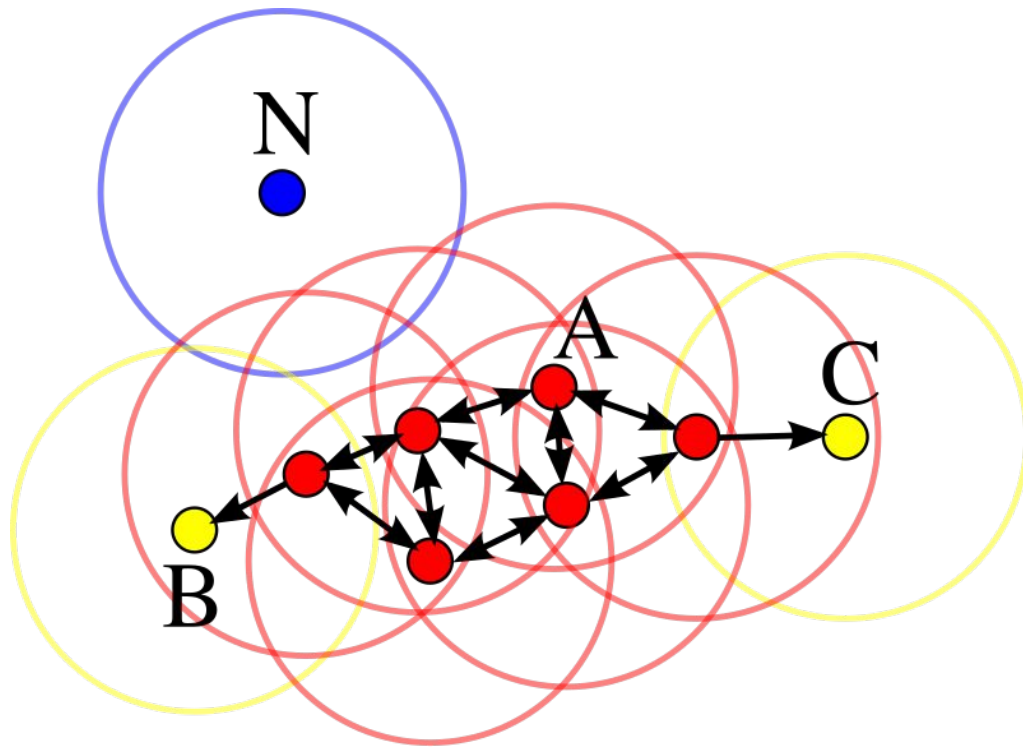
nichePCA

Schaub & Yousefi et al. 2025

<https://academic.oup.com/bioinformatics/article/41/1/btaf005/7945104>

Figure modified from

<https://www.biorxiv.org/content/10.1101/2024.12.18.629206v1.full>



DBSCAN

Ester et al., 1996

Image source: <https://en.wikipedia.org/wiki/DBSCAN>

Questions?